

Proyecto final: Cardiopulmonary Physiological Monitoring System

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Descripción

Es una aplicación móvil diseñada para observar y analizar cómo reacciona el cuerpo de una persona durante el ejercicio (como al correr 60 minutos en una caminadora).

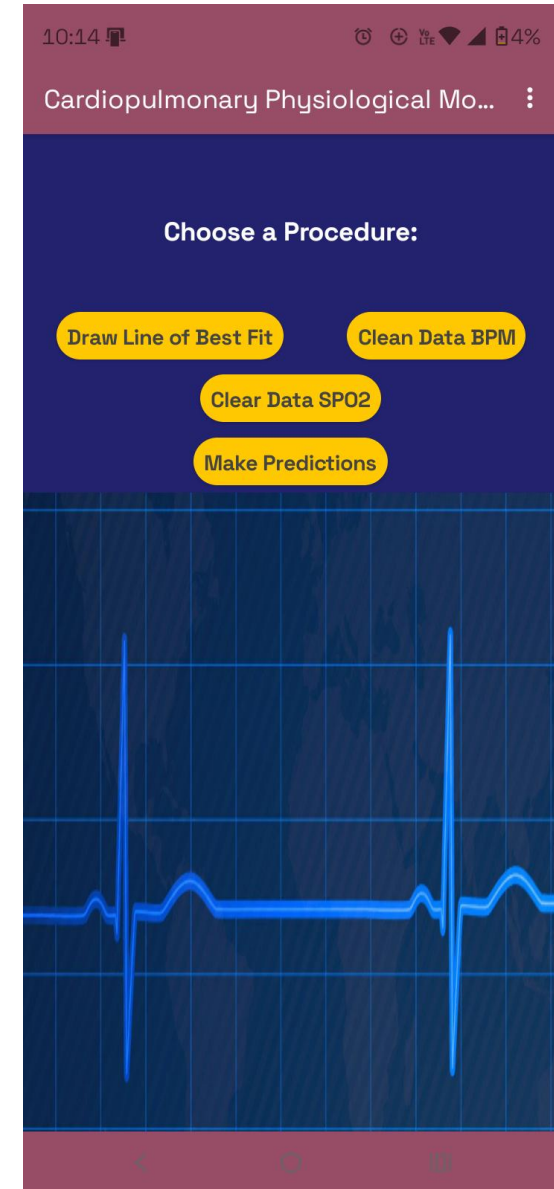
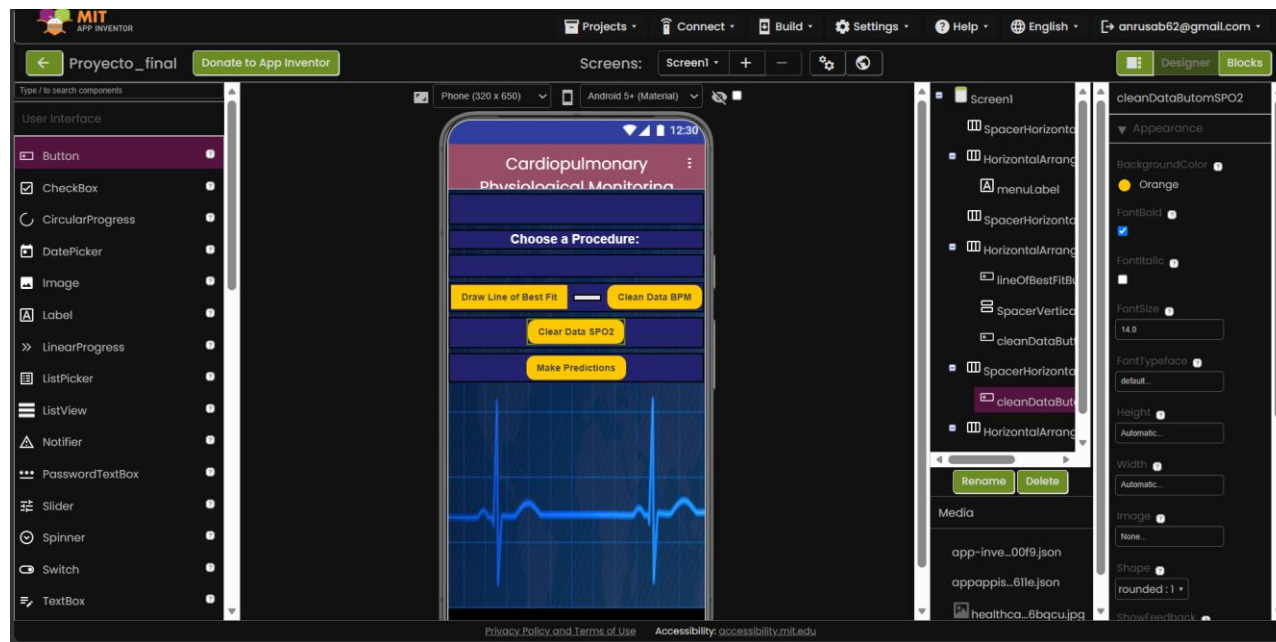
Sus funciones principales son:

- **Carga de datos en la nube:** Importa los registros del ritmo cardíaco y la oxigenación (SpO2) directamente desde Google Sheets.
- **Gráficas visuales:** Dibuja el comportamiento de los signos vitales a lo largo del entrenamiento para verlos de forma clara en la pantalla.
- **Tendencias matemáticas:** Calcula automáticamente la línea de tendencia de los datos para mostrar si el nivel de esfuerzo va en aumento o va bajando.
- **Asistente Inteligente:** Utiliza inteligencia artificial para analizar la rutina, identificar las fases del ejercicio (calentamiento, trabajo y enfriamiento) y explicarte si los resultados son normales.

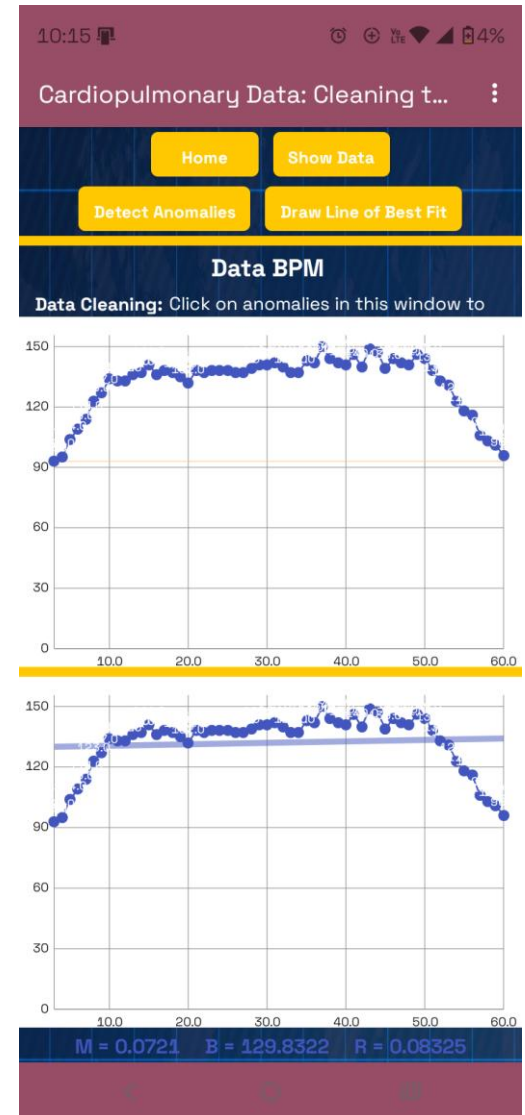
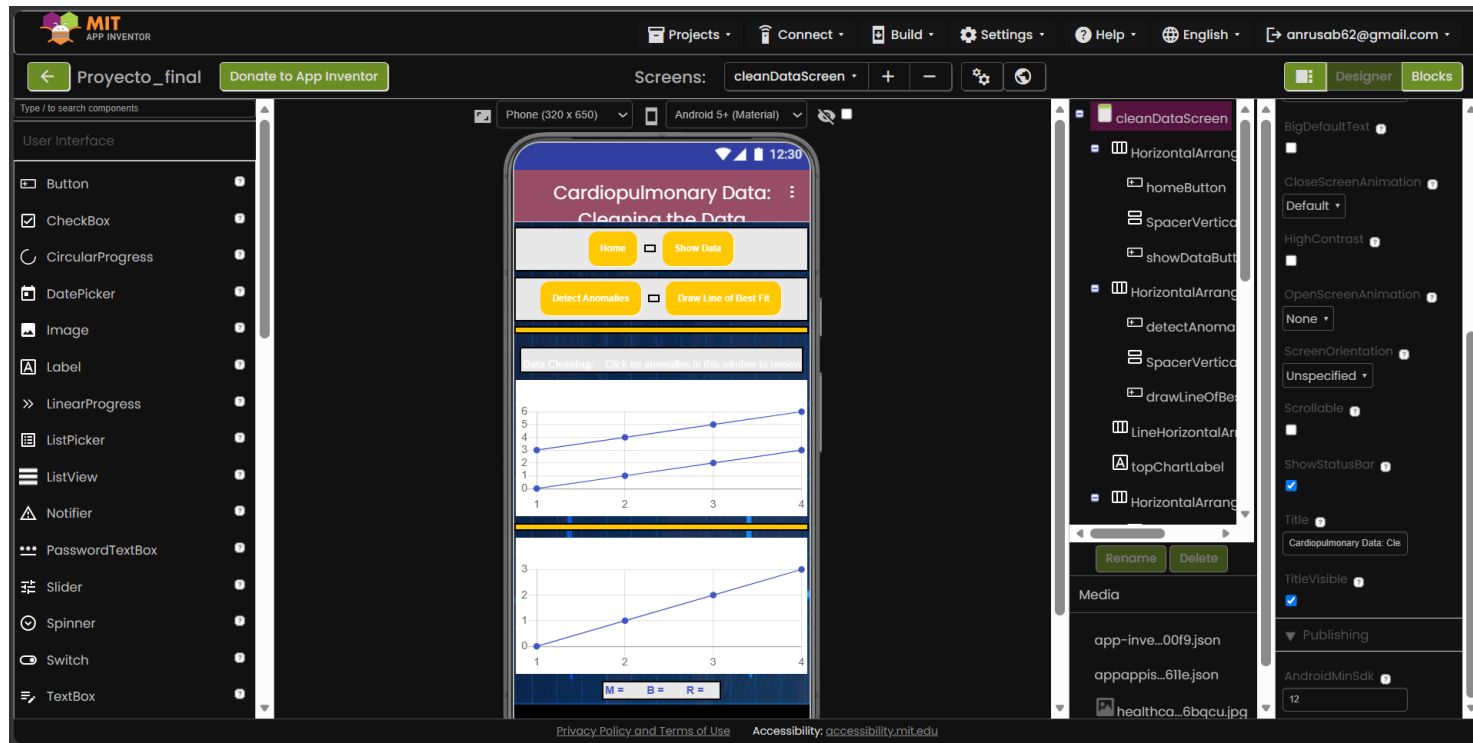
Link de Google Sheets

https://docs.google.com/spreadsheets/d/1o8ahS6B9PD17ydqXQZir2Qz4XQJN_zR6clU6S77rD0k/edit?usp=sharing

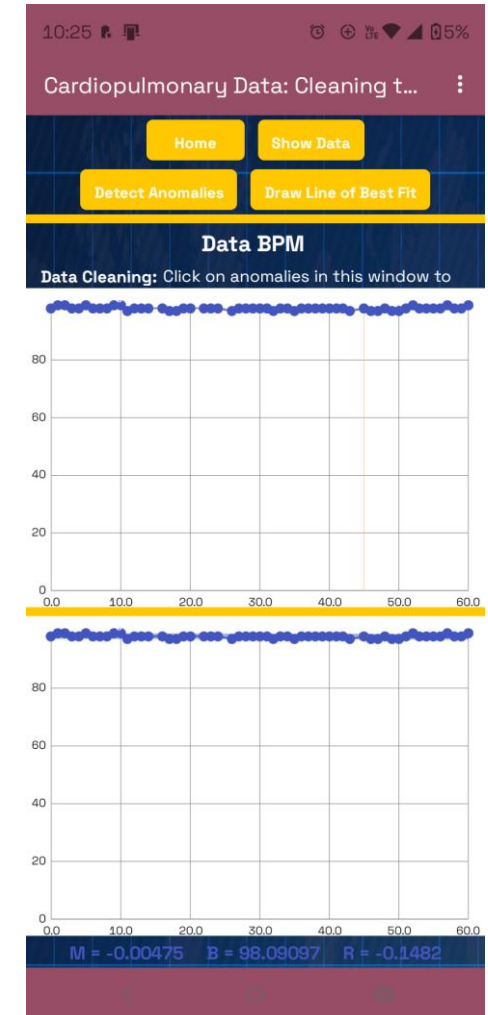
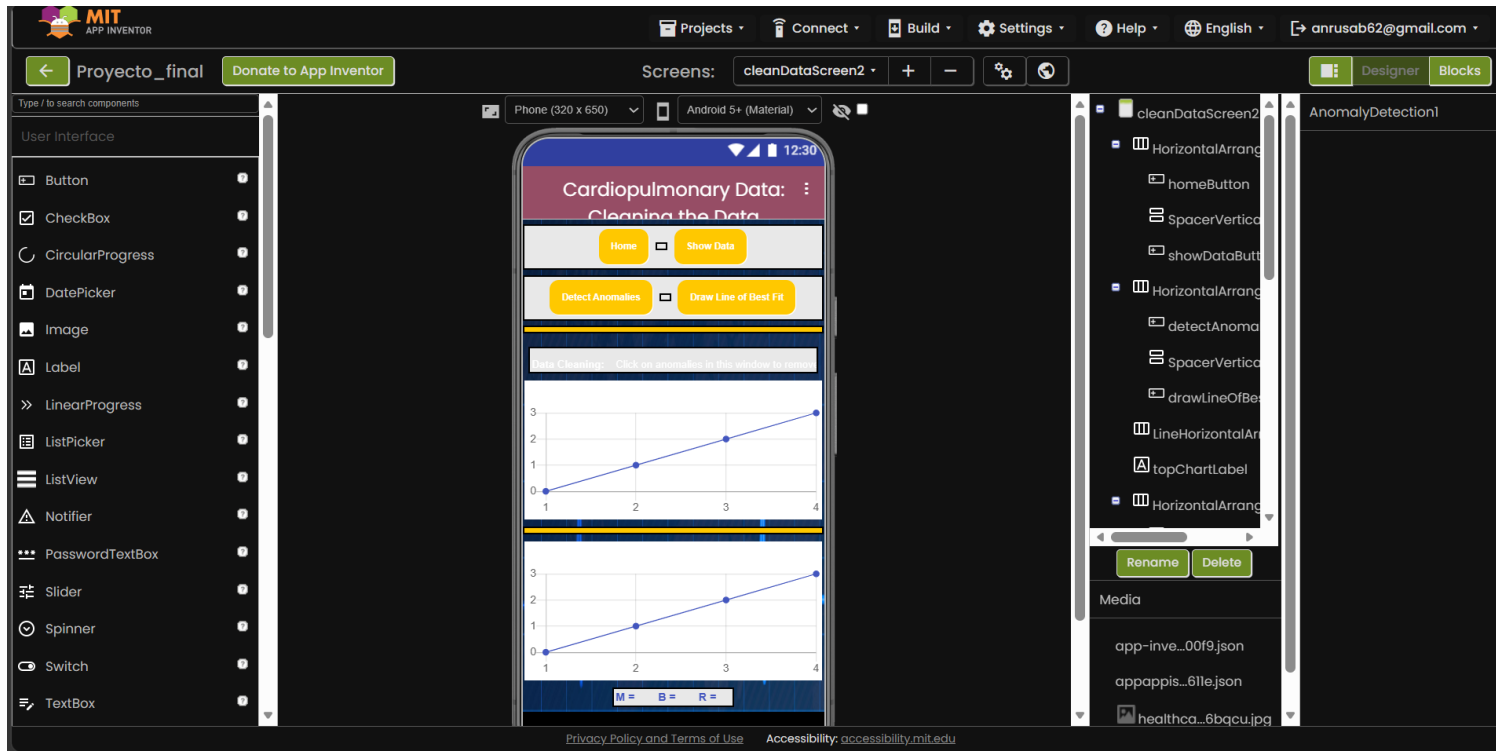
Interfaz



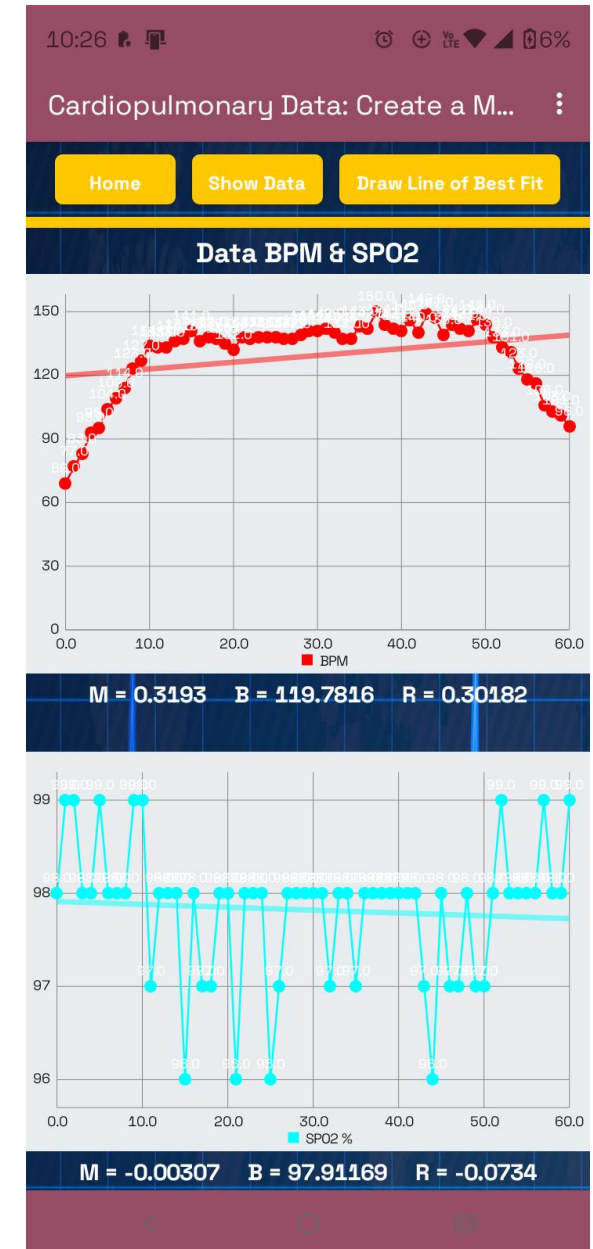
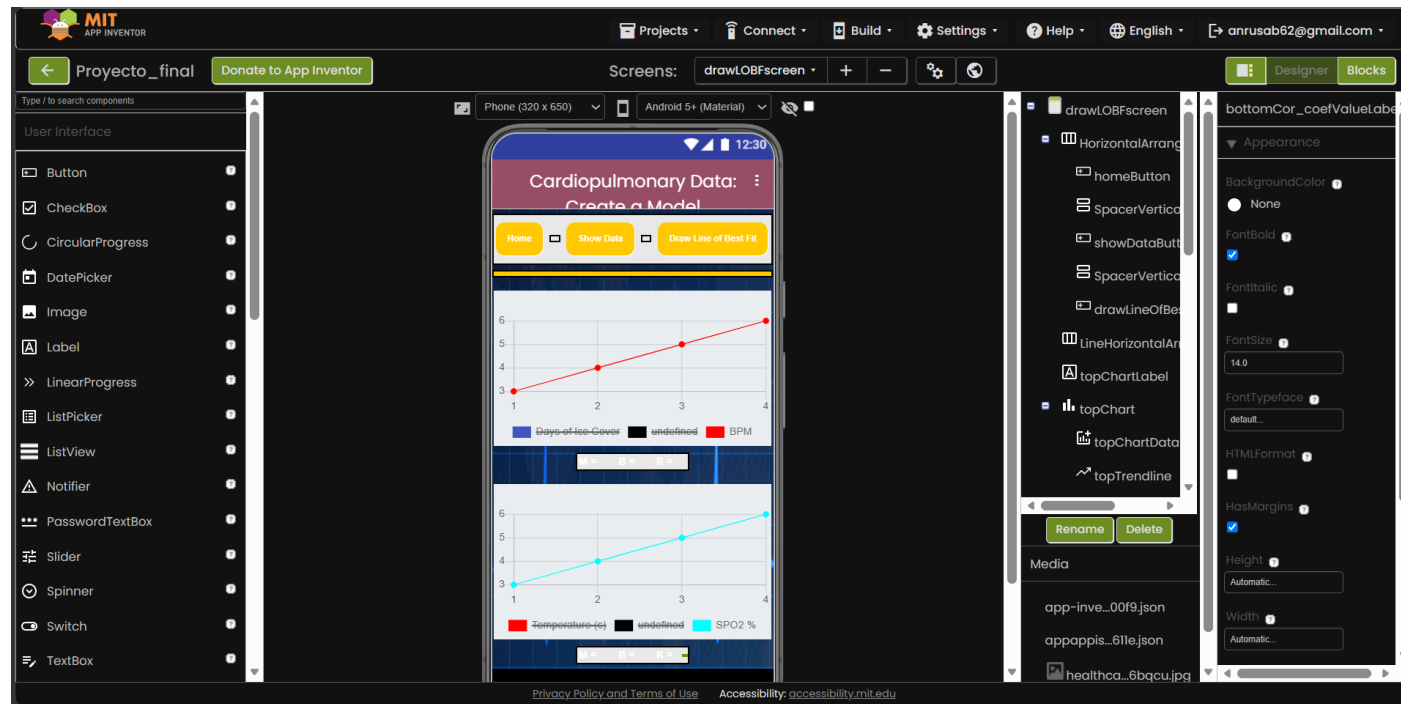
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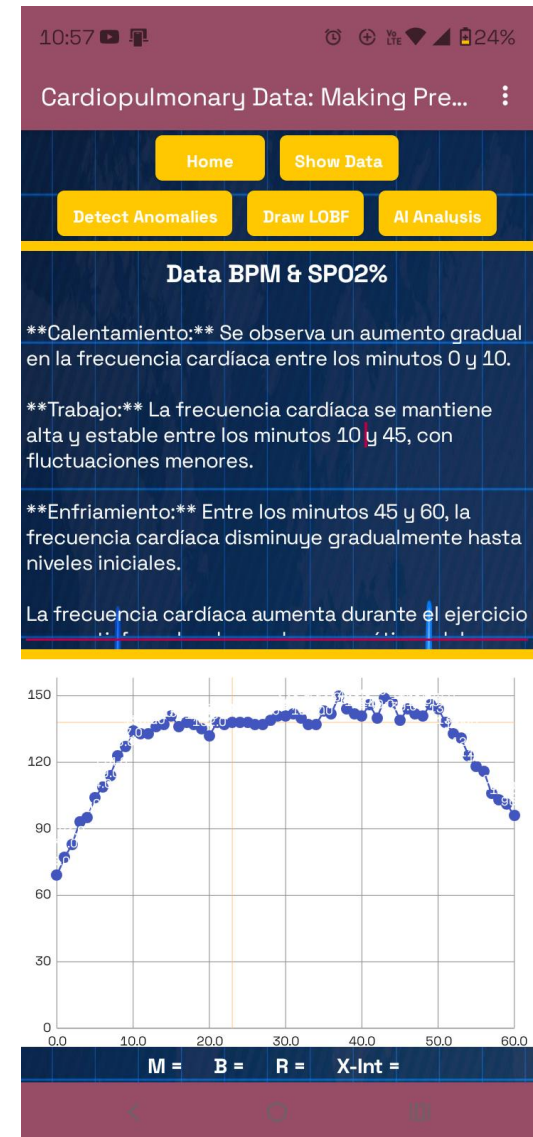
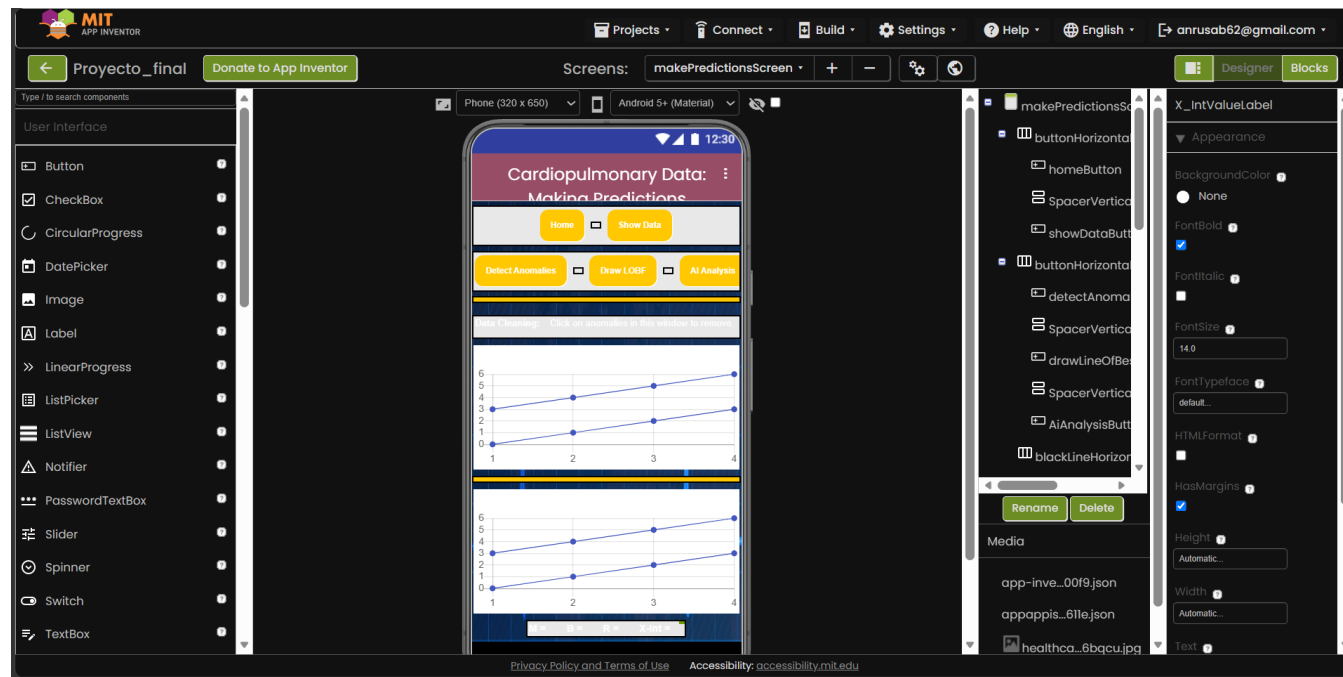
Interfaz



Interfaz



Interfaz



Código

The screenshot displays the MIT App Inventor web interface. The top navigation bar includes the MIT App Inventor logo, a 'Donate to App Inventor' button, and a menu with options: Projects, Connect, Build, Settings, Help, English, and a user profile for anrusab62@gmail.com. Below this, the project name 'Proyecto_final' is shown with a back arrow and a 'Screens:' section containing 'Screen1' and controls for adding or removing screens. The left sidebar features a 'Built-in' category with various block types: Control, Logic, Math, Matrices, Text, Lists, Dictionaries, Colors, Variables, Procedures, and a 'Screen1' block. Below these are 'Rename' and 'Delete' buttons, followed by a 'Media' section listing files like 'app-inve...00f9.json', 'apppapis...61le.json', and 'healthca...6bqcu.jpg'. The main workspace is a code editor with a dark background, showing four event-driven code blocks. Each block starts with a 'when' trigger (button click) followed by a 'do' action (open another screen). The blocks are: 1. 'when lineOfBestFitButton.Click' leading to 'do open another screen screenName drawLOBFscreen'. 2. 'when cleanDataButtonBPM.Click' leading to 'do open another screen screenName cleanDataScreen'. 3. 'when cleanDataButomSPO2.Click' leading to 'do open another screen screenName cleanDataScreen2'. 4. 'when makePredictionsButton.Click' leading to 'do open another screen screenName makePredictionsScreen'. On the right side of the code editor, there are icons for a backpack, a checklist, a target, zoom in (+) and zoom out (-) buttons, and a trash can. At the bottom of the interface, there are links for 'Privacy Policy and Terms of Use' and 'Accessibility: accessibility.mit.edu'.

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Proyecto_final Donate to App Inventor Screens: Screen1 + -

Built-in

- Control
- Logic
- Math
- Matrices
- Text
- Lists
- Dictionaries
- Colors
- Variables
- Procedures
- Screen1

Rename Delete

Media

- app-inve...00f9.json
- apppapis...61le.json
- healthca...6bqcu.jpg

Show Warnings

when lineOfBestFitButton.Click

do open another screen screenName drawLOBFscreen

when cleanDataButtonBPM.Click

do open another screen screenName cleanDataScreen

when cleanDataButomSPO2.Click

do open another screen screenName cleanDataScreen2

when makePredictionsButton.Click

do open another screen screenName makePredictionsScreen

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Código

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Proyecto_final Donate to App Inventor

Screens: cleanDataScreen + -

Designer Blocks

Built-in

- Control
- Logic
- Math
- Matrices
- Text
- Lists
- Dictionaries
- Colors
- Variables
- Procedures
- cleanDataScreen

Media

- app-inve...00f9.json
- appappis...611e.json
- healthca...6bqcu.jpg

when homeButton Click

- do open another screen screenName Screen1

when showDataButton Click

- do call cleanedChartData2D Clear
- call dataCleaningChartData2D Clear
- call spreadsheet1 ReadSheet sheetName Monitoreo_Complet_60m

when spreadsheet1 GotSheetData

- do set topChartLabel Text to Data BPM
- call cleanedChartData2D ImportFromSpreadsheet spreadsheet spreadsheet1 xColumn Tiempo (minutos) yColumn Frecuencia Cardiaca (BPM) useHeaders true
- call dataCleaningChartData2D ImportFromSpreadsheet spreadsheet spreadsheet1 xColumn Tiempo (minutos) yColumn Frecuencia Cardiaca (BPM) useHeaders true

when drawLineOfBestFitButton Click

- do set Trendline1 ChartData to cleanedChartData2D

when detectAnomaliesButton Click

- do call dataCleaningChartData2D HighlightDataPoints dataPoints dataCleaningChartData2D color red
- call AnomalyDetection1 DetectAnomaliesInChartData chartData dataCleaningChartData2D threshold 2

when dataCleaningChartData2D EntryClick

- do if is in list? thing make a list get x get y list call AnomalyDetection1 DetectAnomaliesInChartData chartData dataCleaningChartData2D threshold 2 then call dataCleaningChartData2D RemoveEntry x get x y get y call cleanedChartData2D Clear call cleanedChartData2D ImportFromList list call dataCleaningChartData2D GetAllEntries

when Trendline1 Updated

- do results set topSlopeValueLabel Text to Trendline1 LinearCoefficient set topY_intValueLabel Text to Trendline1 YIntercept set topCor_coefValueLabel Text to Trendline1 CorrelationCoefficient

Show Warnings

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Código

The screenshot displays the MIT App Inventor web interface for a project named "Proyecto_final". The top navigation bar includes links for Projects, Connect, Build, Settings, Help, and a user profile. The left sidebar lists built-in components: Control, Logic, Math, Matrices, Text, Lists, Dictionaries, Colors, Variables, Procedures, and a media section. The main workspace shows two screens: "cleanDataScreen2" and "detectAnomaliesButton".

Screen: cleanDataScreen2

- when homeButton.Click**
 - do open another screen screenName Screen1
- when showDataButton.Click**
 - do
 - call cleanedChartData2D.Clear
 - call dataCleaningChartData2D.Clear
 - call spreadsheet1.ReadSheet
 - sheetName Monitoreo_Complet_60m
- when spreadsheet1.GotSheetData**
 - do
 - set topChartLabel.Text to Data BPM
 - call cleanedChartData2D.ImportFromSpreadsheet
 - spreadsheet spreadsheet1
 - xColumn Tiempo (minutos)
 - yColumn Saturacion de Oxigeno (%)
 - useHeaders true
 - call dataCleaningChartData2D.ImportFromSpreadsheet
 - spreadsheet spreadsheet1
 - xColumn Tiempo (minutos)
 - yColumn Saturacion de Oxigeno (%)
 - useHeaders true
- when drawLineOfBestFitButton.Click**
 - do set Trendline1.ChartData to cleanedChartData2D

Screen: detectAnomaliesButton

- when detectAnomaliesButton.Click**
 - do call dataCleaningChartData2D.HighlightDataPoints
 - dataPoints call AnomalyDetection1.DetectAnomaliesInChartData
 - chartData dataCleaningChartData2D
 - threshold 2
 - color red
- when dataCleaningChartData2D.EntryClick**
 - do
 - if is in list? thing
 - make a list
 - get X
 - get Y
 - list call AnomalyDetection1.DetectAnomaliesInChartData
 - chartData dataCleaningChartData2D
 - threshold 2
 - then call dataCleaningChartData2D.RemoveEntry
 - x get X
 - y get Y
 - call cleanedChartData2D.Clear
 - call cleanedChartData2D.ImportFromList
 - list call dataCleaningChartData2D.GetAllEntries
- when Trendline1.Updated**
 - do
 - set topSlopeValueLabel.Text to Trendline1.LinearCoefficient
 - set topY_intValueLabel.Text to Trendline1.YIntercept
 - set topCor_coefValueLabel.Text to Trendline1.CorrelationCoefficient

Código

The screenshot displays the MIT App Inventor web interface. The top navigation bar includes links for Projects, Connect, Build, Settings, Help, and a language selector set to English. The user's email address, anrusab62@gmail.com, is also visible. Below the navigation bar, the project name 'Proyecto_final' is shown, along with a 'Donate to App Inventor' button. The 'Screens' section indicates the current screen is 'drawLOBscreen'. The interface is divided into three main panels: a left sidebar with a 'Built-in' components list (Control, Logic, Math, Matrices, Text, Lists, Dictionaries, Colors, Variables, Procedures) and a 'Media' section; a central code editor; and a right sidebar with a 'Designer' tab and a 'Blocks' tab. The code editor contains three event-driven blocks: 1. A 'when homeButton.Click' block with a 'do' block containing 'open another screen' set to 'Screen1'. 2. A 'when showDataButton.Click' block with a 'do' block containing three 'call' blocks: 'topChartData2D.Clear', 'bottomChartData2D.Clear', and 'spreadsheet1.ReadSheet' with 'sheetName' set to 'Monitoreo_Complet_60m'. 3. A 'when spreadsheet1.GotSheetData' block with a 'do' block containing two 'call' blocks: 'topChartData2D.ImportFromSpreadsheet' and 'bottomChartData2D.ImportFromSpreadsheet'. Both 'call' blocks have 'spreadsheet' set to 'spreadsheet1' and 'useHeaders' set to 'true'. The 'top' block has 'xColumn' set to 'Tiempo (minutos)' and 'yColumn' set to 'Frecuencia Cardiaca (BPM)'. The 'bottom' block has 'xColumn' set to 'Tiempo (minutos)' and 'yColumn' set to 'Saturacion de Oxigeno (%)'. A fourth block, 'when drawLineOfBestFitButton.Click', is partially visible on the right, containing a 'do' block with multiple 'set' blocks for 'topTrendline', 'bottomTrendline', and their respective 'LinearCoefficient' and 'YIntercept' values. The bottom of the interface features links for 'Privacy Policy and Terms of Use' and 'Accessibility: accessibility.mit.edu'.

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Proyecto_final Donate to App Inventor

Screens: drawLOBscreen + -

Designer Blocks

Built-in

- Control
- Logic
- Math
- Matrices
- Text
- Lists
- Dictionaries
- Colors
- Variables
- Procedures
- drawLOBscreen

Media

- app-inve...00f9.json
- appappis...611e.json
- healthca...6bqcu.jpg

when homeButton.Click

do

- open another screen screenName Screen1

when showDataButton.Click

do

- call topChartData2D.Clear
- call bottomChartData2D.Clear
- call spreadsheet1.ReadSheet sheetName Monitoreo_Complet_60m

when spreadsheet1.GotSheetData

do

- set topChartLabel.Text to Data BPM & SPO2
- call topChartData2D.ImportFromSpreadsheet spreadsheet spreadsheet1 xColumn Tiempo (minutos) yColumn Frecuencia Cardiaca (BPM) useHeaders true
- call bottomChartData2D.ImportFromSpreadsheet spreadsheet spreadsheet1 xColumn Tiempo (minutos) yColumn Saturacion de Oxigeno (%) useHeaders true

when drawLineOfBestFitButton.Click

do

- set topTrendline.ChartData to topChartData2D
- set bottomTrendline.ChartData to bottomChartData2D
- set topSlopeValueLabel.Text to topTrendline.LinearCoefficient
- set topY_intValueLabel.Text to topTrendline.YIntercept
- set topCor_coefValueLabel.Text to topTrendline.CorrelationCoefficient
- set bottomSlopeValueLabel.Text to bottomTrendline.LinearCoefficient
- set bottomY_intValueLabel.Text to bottomTrendline.YIntercept
- set bottomCor_coefValueLabel.Text to bottomTrendline.CorrelationCoefficient

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Código

